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# 1 GLM Overview

## 1.1 Example

Normal Regression:  $y_i = \beta_0 + \beta_1 x_i + \epsilon_i$ :  $\epsilon_i \stackrel{\text{iid}}{\sim} N(0, \sigma^2)$

Logistic Regression:  $y_i \stackrel{\text{ind}}{\sim} \text{Bern}(p_i)$ :  $\text{logit}(p_i) = \beta_0 + \beta_1 x_i$ ,  $\log(\frac{p_i}{1-p_i})$

Differences:

- Support of  $y_i$ :  $y_i \in \mathbb{R}$  vs.  $y_i \in \{0, 1\}$
- Link functions:  $g(\mu_i) = \mu_i$  vs.  $g(p_i) = \log(\frac{p_i}{1-p_i})$
- Likelihoods:  $y_i \stackrel{\text{ind}}{\sim} N(\mu_i, \sigma^2)$  vs.  $y_i \stackrel{\text{ind}}{\sim} \text{Bern}(p_i)$

Similarities:

- Univariate models
- Exponential families
- Independently distributed
- Linear (or otherwise systematic) relationship with a function of  $E(y_i)$
- Have link functions
- Assuming that  $y_i$  is random
- $\beta$ 's are unknown

## 1.2 GLMs

Generalized Linear Model:

- Random component:  $Y_i \sim f(\mu_i)$
- Link component:  $g(\mu_i) = \eta_i$
- Systematic component:  $\eta_i = x_i^\top \beta$

We generally use GLMs for the following:

- Inference: understand relationships
- Prediction: behavior of specific  $x$ 's